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Q1. Intelligent Customer Feedback Analysis

CODE

```
def preprocess(text):
    text=text.lower()
    text=text.strip()
    text=text.replace(".", "")
    text=text.replace(",", "")
    text=text.replace("!", "")
    text=text.replace("?", "")
    return text

def analyze(text):
    positive=["good", "excellent", "great", "happy", "fast"]
    negative=["bad", "poor", "slow", "worst", "late"]
    pcount=0
    ncount=0
    words=text.split()
    for word in words:
        if word in positive:
            pcount+=1
        elif word in negative:
            ncount+=1
    if pcount>ncount:
        result="positive"
    elif ncount>pcount:
        result="negative"
    else:
        result="netural"
    print("positive words:", pcount)
    print("negative words:", ncount)
    print("feedback:", result)
n=int(input("enter no of feedbacks:"))
for i in range(n):
    print("\nfeedback", i+1)
    feedback= input("enter feedback: ")
    clean=preprocess(feedback)
    analyze(clean)
```

INPUT

```
3
excellent service! very good.
bad delivery and worst support.
good but late delivery.
```

OUTPUT

```
enter no of feedbacks:
feedback 1
enter feedback: positive words: 2
negative words: 0
feedback: positive

feedback 2
enter feedback: positive words: 0
negative words: 2
feedback: negative

feedback 3
enter feedback: positive words: 1
negative words: 1
feedback: netural
```

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Q2. Comprehensive Password Strength Checker

CODE

```
password=input()
upper=0
lower=0
digit=0
symbol=0
for ch in password:
    if 'A' ≤ ch ≤ 'Z':
        upper+=1
    elif 'a' ≤ ch ≤ 'z':
        lower+=1
    elif '0' ≤ ch ≤ '9':
        digit+=1
    else:
        symbol+=1
score=0
if len(password) ≥ 8:
    score+=1
if upper>0:
    score+=1
if lower>0:
    score+=1
if digit>0:
    score+=1
if symbol>0:
    score+=1
if score ≤ 2:
    print("weak")
elif score==3:
    print("moderate")
elif score==4:
    print("strong")
else:
    print("very strong")
```

INPUT OUTPUT

Q3. Structured Server Log Error Analyser**CODE**

```
def analyze_logs(logs):
    unique_errors=[]
    error_count=0
    print("error messages:")
    for log in logs:
        if "error" in log.lower() or log.lower().startswith("error") or log.lower().find("error")!=-1:
            error_count+=1
            if log not in unique_errors:
                unique_errors.append(log)
                print(log)
    print("total errors:", error_count)
    print("unique errors:", len(unique_errors))

n=int(input())
logs=[]
for i in range(n):
    logs.append(input())
analyze_logs(logs)
```

INPUT

```
5
INFO: Login Successful
ERROR: File Not Found
WARNING: Low Memory
ERROR: File Not Found
ERROR: Database Failed
```

OUTPUT

```
error messages:
ERROR: File Not Found
ERROR: Database Failed
total errors: 3
unique errors: 2
```

Q4. Email Domain Frequency Analyser

CODE

```
n=int(input())
domains={}
for i in range(n):
    email=input()
    if '@' in email:
        parts=email.split("@")
        if len(parts)==2 and parts[1]!="":
            domain=parts[1]
            if domain in domains:
                domains[domain]+=1
            else:
                domains[domain]=1
        else:
            print("invalid email")
    else:
        print("invalid email")
max_count=0
max_domain=""
for domain in domains:
    print(domain,domains[domain])
    if domains[domain]>max_count:
        max_count=domains[domain]
        max_domain=domain
print("most used domain:", max_domain)
print("frequency:", max_count)
```

INPUT

```
5
abc@gmail.com
xyz@yahoo.com
prq@gmail.com
hello@gmail.com
test@yahoo.com
```

OUTPUT

```
gmail.com 3
yahoo.com 2
most used domain: gmail.com
frequency: 3
```